

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

TECHNICAL SAFETY DIRECTIVE: SOP-CDU-SAFE-014 (REV 3)

SUBJECT: Process Piping Isolation, Relief Device Sizing and SIF Prevention

1.0 Purpose and Statutory Scope

This procedure applies to all crude distillation and residue processing units across MRPL Phase I, II, and III facilities. It incorporates mandatory compliances under API 570 and ASME B31.3.

2.0 Thermal Expansion Overpressure Protection (ASME B31.3 §301.5.1)

Where liquid can be blocked in by closure of valves and subsequent heating by solar radiation or process heat transfer can occur, overpressure protection shall be provided.

Double-blocked liquid circuits exceeding 120 deg C must incorporate a certified Thermal Relief Valve (TRV) sized for minimum 110% MAWP, discharging to closed blowdown.

3.0 Relief Device Isolation and Mechanical Interlocks (API 570 §5.7)

All isolation valves upstream and downstream of pressure safety valves (PSVs) shall be car-sealed open (CSO). Bypass valves around relief valves must be car-sealed closed (CSC) and locked.

4.0 Wall Thickness Retirement Limits and Ultrasonic Scanning Cycles

When remaining service life is computed to be less than 3.0 years, inspection frequency must be doubled. Continuous ultrasonic thickness logging is required for circuits with corrosion rates > 0.3mm/yr.